

# AA1 – Proof by induction

## (solutions)

1. It is enough to prove the equality of two sets, A and B, by proving by induction that all elements of A pertain to B. 

☐ TRUE

☒ FALSE

2. A single counter example is enough to prove the falsity of a statement?

☒ YES

☐ NO

☐ YES, but we have to present the counter example and still show that the proof by induction is not possible.

3. An integer  $n$  is a prime if  $n > 1$  and if the only divisors of  $n$  are 1 and  $n$ . It is possible to prove that all numbers are either a prime number or a product of primes?

☒ YES

☐ YES, but it is not possible to prove by induction.

☐ NO

4. To programmatically generate prime numbers, one can use the formula:  $p = 6 * n \pm 1$ , for  $n \geq 1$ . Which of the following sentences is true?

☐ It is possible to proof by induction that all numbers generated by the formula are prime numbers.

☒ It is possible to proof by induction that all prime numbers over 3 are given by this formula.

☐ We cannot proof by induction as there is no possible inductive definition.

5. For the base case we can use any element of the inductive structure.

☐ TRUE

☒ FALSE

6. It is possible to prove by induction an hypothesis with more than one variable, however...

- ☐ We only need to do the inductive step in one variable.
- ☒ We need to do the inductive step for all variables.

7. Is it possible to prove by induction that  $11n-6$  is a multiple of 5 for  $n>0$ ? 

- ☐ YES
- ☒ YES, but we need to use the inventor's paradox for a stronger proof.
- ☐ NO